



Motivation

Distilled Dataset s.t. **Diversity** + **Generalization** + **Representativeness**

We seek to study:

- Do the priors in vanilla Diffusion Models (DMs) satisfy the requirements for Dataset Distillation (DD)?
- Are there unused priors in DMs that could benefit DD?

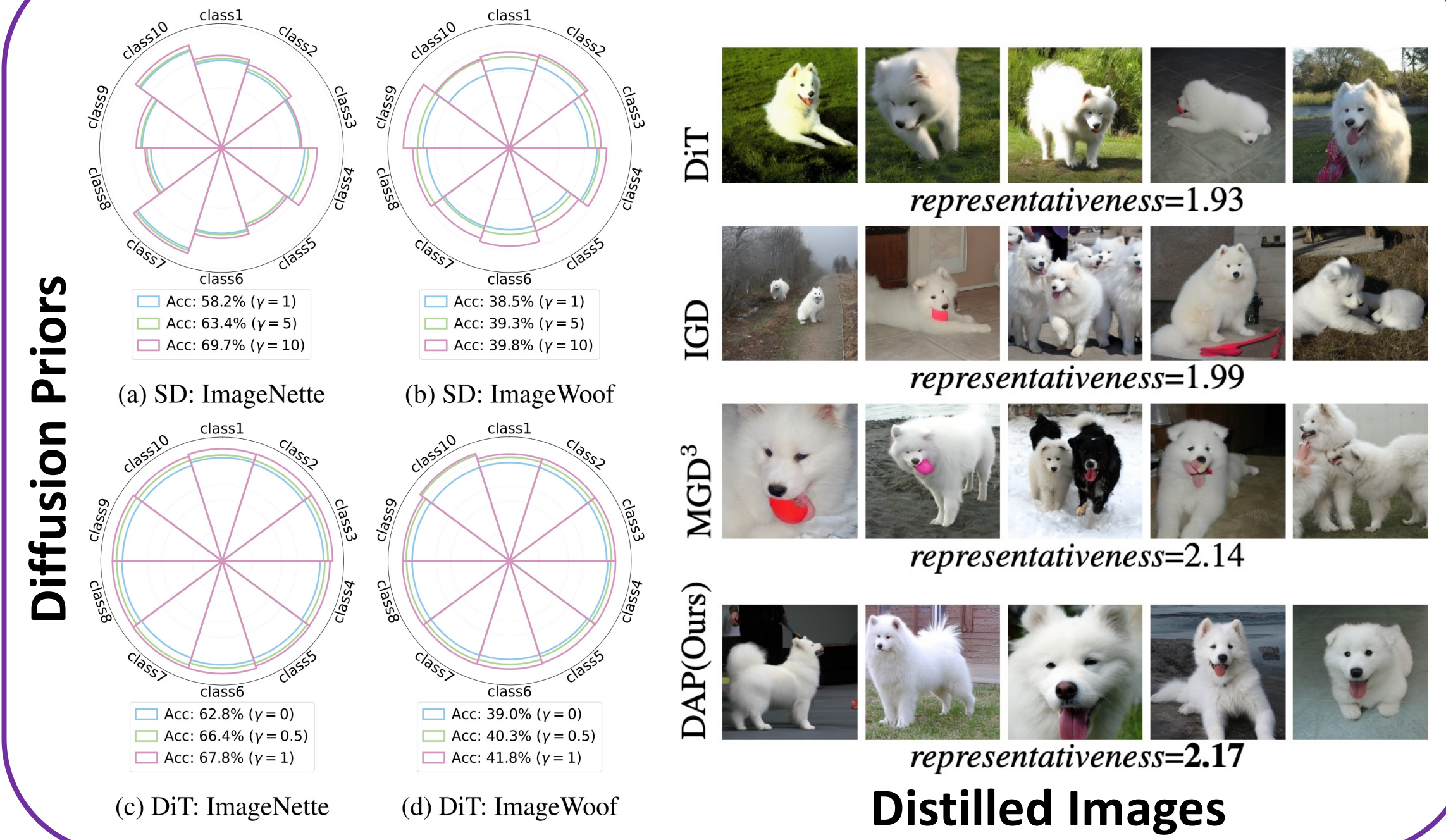
Our Contribution:

- We prove the priors in the well-trained DMs meet the diversity and generalization requirements of DD.
- We derive the overlooked **Representativeness Priors** from DMs and formalize it into a kernel-induced distance, which guides the sampling dynamic and improves the quality of distilled datasets.

Table 1: NLLs ↓ on different datasets. The results are computed by a vanilla diffusion model (Ho et al., 2020) trained on ImageNet.

Dataset	Training Set	Test Set
ImageNette	2.4452±1.03	2.6327±1.08
ImageWoof	2.5856±0.88	3.0838±0.86

Visualizations



DAP: Diffusion As Priors

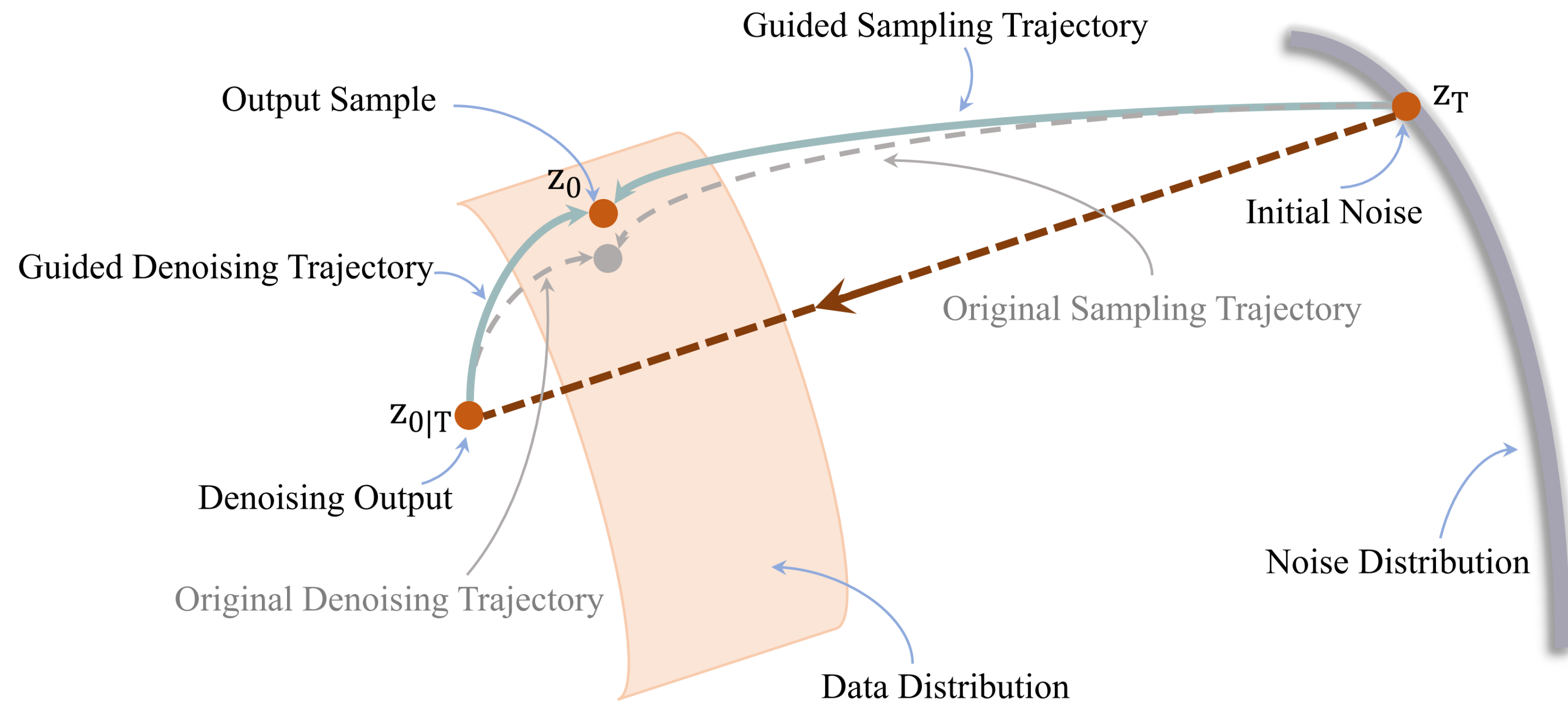
$$\nabla_{\mathbf{x}} \log p(\mathbf{x}|\mathcal{R}) = \underbrace{\nabla_{\mathbf{x}} \log p(\mathbf{x})}_{\text{Diversity \& Generalization}} + \underbrace{\nabla_{\mathbf{x}} \log p(\mathcal{R}|\mathbf{x})}_{\text{Representativeness}}.$$

Prior Measurement

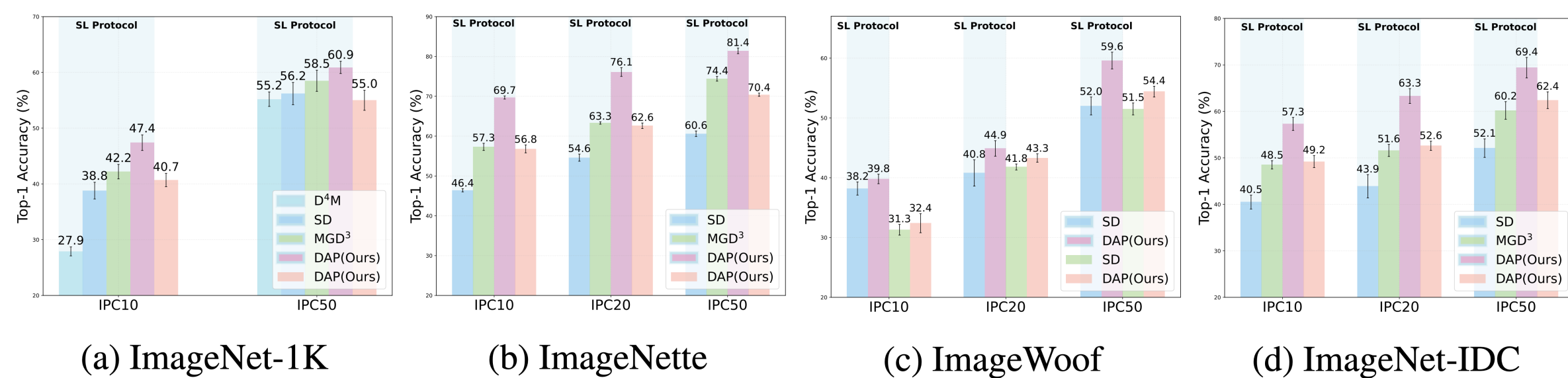
$$\begin{aligned} \nabla_{\mathbf{x}^{syn}} \log p(\mathcal{R}|\mathbf{x}^{syn}) &= \nabla_{\mathbf{x}^{syn}} \log \frac{\{\exp[-\frac{1}{N} \sum_N \mathcal{D}_{\mathcal{K}}(\mathbf{x}^{syn}, \mathbf{x}^{train})]\}^\gamma}{Z} \\ &= \nabla_{\mathbf{x}^{syn}} \log \frac{\{\exp[-\frac{1}{N} \sum_N d(\phi(\mathbf{x}^{syn}), \phi(\mathbf{x}^{train}))]\}^\gamma}{Z} \\ &\propto -\gamma \frac{1}{N} \sum_N \nabla_{\mathbf{x}^{syn}} d(\phi(\mathbf{x}^{syn}), \phi(\mathbf{x}^{train})), \end{aligned}$$

Prior Guidance

$$\begin{aligned} d\mathbf{x} &= [f(\mathbf{x}_t^{syn}, t) - g(t)^2 (\nabla_{\mathbf{x}_t^{syn}} \log p(\mathbf{x}_t^{syn}) + \gamma \nabla_{\mathbf{x}_t^{syn}} \log p(\mathcal{R}|\mathbf{x}_t^{syn}))] dt + g(t) d\mathbf{w} \\ &\propto [f(\mathbf{x}_t^{syn}, t) - g(t)^2 (-\epsilon_\theta(\mathbf{x}_t^{syn}, t) + \gamma \nabla_{\mathbf{x}_t^{syn}} d(\phi(\mathbf{x}_t^{syn}), \phi(\mathbf{x}_t^{train})))] dt + g(t) d\mathbf{w}, \end{aligned}$$



Results On Stable Diffusion



Conclusion

- We identify **Diversity**, **Generalization**, and **Representativeness** priors in diffusion models.
- We posit the **Representativeness** prior is unused in previous works.
- We demonstrate how these three priors can be integrated to guide the generation process.
- DAP works well with both hard-label and soft-label evaluation protocol.
- DAP is compatible with common used diffusers library, just enjoy it!